

A

B

C

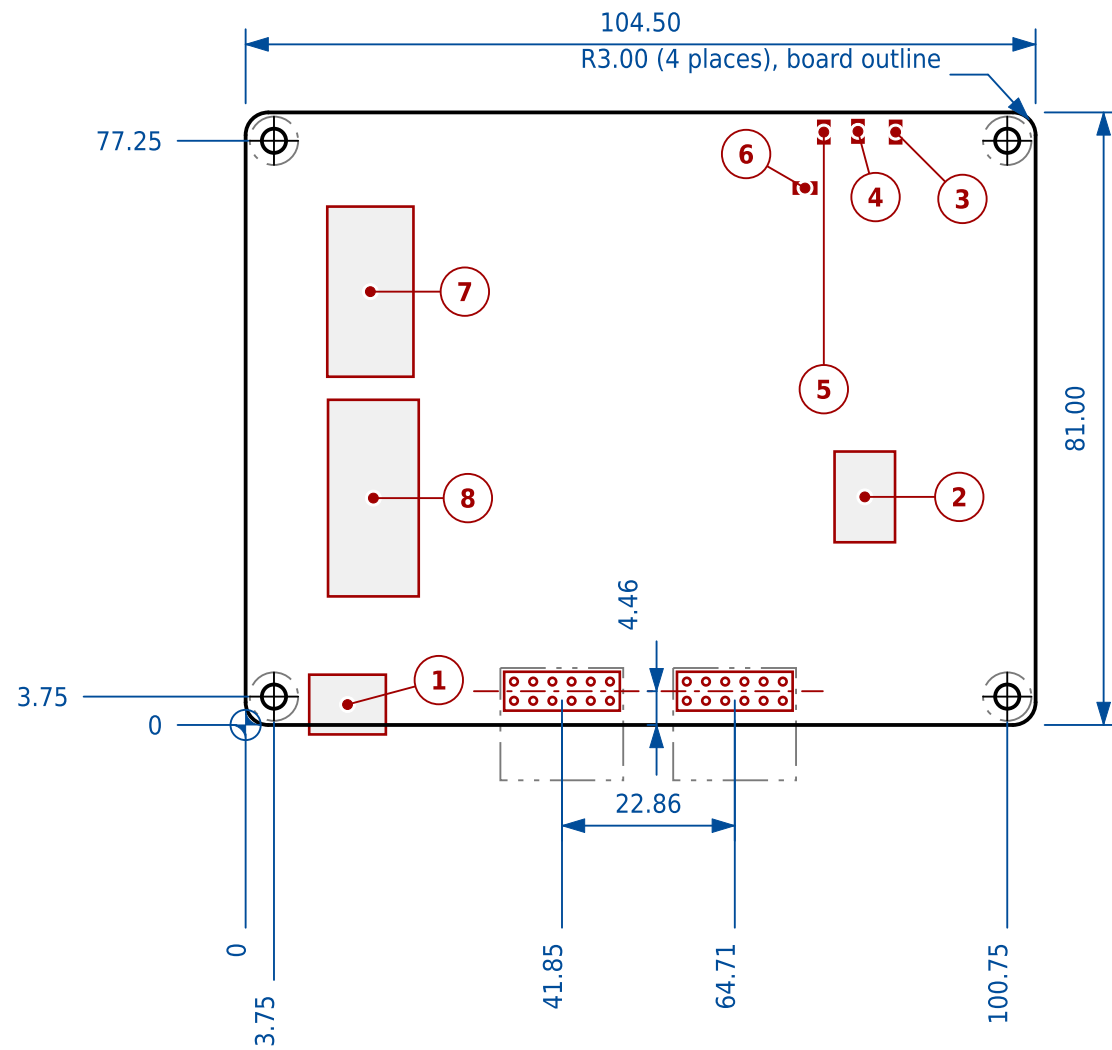
D

A

B

C

D



## NOTES

- All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
- Geometry is design nominal, read from the KiCad board file.
- Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
- The title block reads "TinyTapeout 1,2,3 Demo Board", but no PCB served TT01: that was a bare-die trial run.
- This sheet covers revisions 2.2.5 and 2.2.6: the same board mechanically, compared feature by feature before merging. Only the electrical design and the shuttle differ.
- The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0, is the same on every revision; their distance from the lower edge and their position along it are not.
- Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
- Board thickness is 1.6 mm nominal. The KiCad file's 1.60000 mm is its stackup sum, not a specified finished thickness.

## NOTES (continued)

- Assembled envelope, connector overhang included, is 104.50 x 82.25 mm against a 104.50 x 81.00 mm board outline. The FEATURE SCHEDULE gives the extents.
- GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
- Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

## SOURCES

- KiCad board file:  
<https://github.com/TinyTapeout/tt123-demo-pcb>  
mpw-mb1.kicad\_pcb @ 303509a - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.5, dated 2023-11-13
- KiCad board file:  
<https://github.com/TinyTapeout/tt123-demo-pcb>  
mpw-mb1.kicad\_pcb @ 3d721a6 - title block: TinyTapeout 1,2,3 Demo Board rev 2.2.6, dated 2024-02-14
- Shuttle mapping: [https://docs.google.com/spreadsheets/d/1xD\\_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370](https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMDqAnrOJLetM/edit?gid=1250963370) - used by: TT02, TT03

## HOLE SCHEDULE

| ID  | X mm   | Y mm  | DIA mm | KEEPOUT mm |
|-----|--------|-------|--------|------------|
| MT1 | 3.75   | 3.75  | 3.20   | 6.40       |
| MT2 | 100.75 | 3.75  | 3.20   | 6.40       |
| MT3 | 3.75   | 77.25 | 3.20   | 6.40       |
| MT4 | 100.75 | 77.25 | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| PMOD A | J3  | bottom | 41.850 | 4.465 | 48.200    | 5.735     |
| PMOD B | J9  | bottom | 64.710 | 4.465 | 71.060    | 5.735     |

## FEATURE SCHEDULE

| #  | FEATURE                                        | X EXTENT mm    | Y EXTENT mm    |
|----|------------------------------------------------|----------------|----------------|
| 1  | USB-C power / control                          | 8.41 to 18.55  | -1.25 to 6.65  |
| 2  | 7-segment display                              | 77.91 to 85.91 | 24.16 to 36.16 |
| 3  | LED D1                                         | 85.25 to 86.72 | 76.92 to 79.88 |
| 4  | LED D2                                         | 80.27 to 81.73 | 77.02 to 79.98 |
| 5  | LED D3                                         | 75.75 to 77.22 | 76.92 to 79.88 |
| 6  | LED D4                                         | 72.52 to 75.48 | 70.27 to 71.73 |
| 7  | 8-way input DIP switch                         | 10.80 to 22.20 | 46.05 to 68.55 |
| 8  | 9-way DIP switch SW2                           | 10.90 to 22.90 | 17.00 to 43.00 |
| 9  | Side Pmod, first position - not on this board  | -              | -              |
| 10 | Side Pmod, second position - not on this board | -              | -              |
| 11 | Side Pmod, third position - not on this board  | -              | -              |

## LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

## DB MPW V2.2.5 / DB MPW V2.2.6

mpw-mb1 rev 2.2.5 and 2.2.6

|                  |                  |                                                |       |
|------------------|------------------|------------------------------------------------|-------|
| DRAWN            | VERSION          | UNITS                                          | SCALE |
| generated        | v0.0-93-g0b19cee | mm                                             | 1:1   |
| SIZE             | SHEET            | DRAWING NO                                     | REV   |
| A3               | 1 OF 1           | TT-DB-01                                       | A     |
| MATERIAL         |                  | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                  | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |